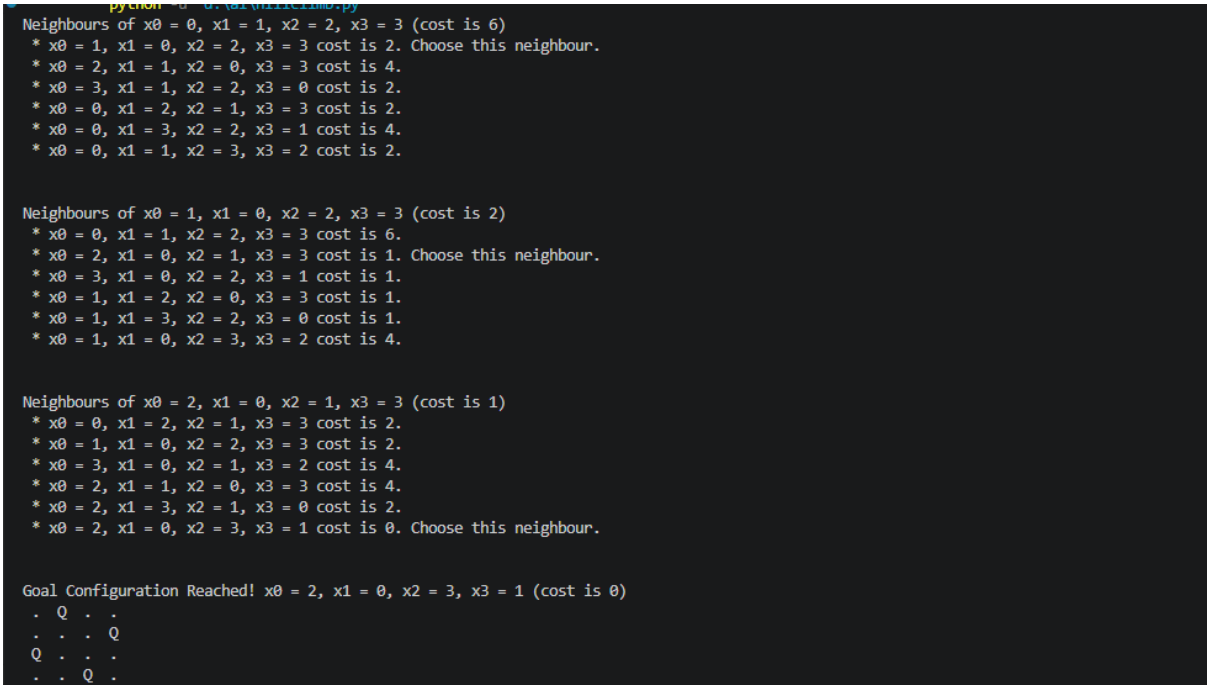
1BM24CS315 WEEK-04

1. Implement Hill Climbing algorithm for 4 queens.

Output:



```
Neighbours of x0 = 0, x1 = 1, x2 = 2, x3 = 3 (cost is 6)
 * x0 = 1, x1 = 0, x2 = 2, x3 = 3 cost is 2. Choose this neighbour.
 * x0 = 2, x1 = 1, x2 = 0, x3 = 3 cost is 4.
 * x0 = 3, x1 = 1, x2 = 2, x3 = 0 cost is 2.
 * x0 = 0, x1 = 2, x2 = 1, x3 = 3 cost is 2.
 * x0 = 0, x1 = 3, x2 = 2, x3 = 1 cost is 4.
 * x0 = 0, x1 = 1, x2 = 3, x3 = 2 cost is 2.


Neighbours of x0 = 1, x1 = 0, x2 = 2, x3 = 3 (cost is 2)
 * x0 = 0, x1 = 1, x2 = 2, x3 = 3 cost is 6.
 * x0 = 2, x1 = 0, x2 = 1, x3 = 3 cost is 1. Choose this neighbour.
 * x0 = 3, x1 = 0, x2 = 2, x3 = 1 cost is 1.
 * x0 = 1, x1 = 2, x2 = 0, x3 = 3 cost is 1.
 * x0 = 1, x1 = 3, x2 = 2, x3 = 0 cost is 1.
 * x0 = 1, x1 = 0, x2 = 3, x3 = 2 cost is 4.


Neighbours of x0 = 2, x1 = 0, x2 = 1, x3 = 3 (cost is 1)
 * x0 = 0, x1 = 2, x2 = 1, x3 = 3 cost is 2.
 * x0 = 1, x1 = 0, x2 = 2, x3 = 3 cost is 2.
 * x0 = 3, x1 = 0, x2 = 1, x3 = 2 cost is 4.
 * x0 = 2, x1 = 1, x2 = 0, x3 = 3 cost is 4.
 * x0 = 2, x1 = 3, x2 = 1, x3 = 0 cost is 2.
 * x0 = 2, x1 = 0, x2 = 3, x3 = 1 cost is 0. Choose this neighbour.


Goal Configuration Reached! x0 = 2, x1 = 0, x2 = 3, x3 = 1 (cost is 0)
 .  Q  .  .
 .  .  .  Q
 Q  .  .  .
 .  .  Q  .
```